TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 1

5-MINUTE REVISION

Basics of Statistics

The building blocks: data, variables, scales and frequency distributions.

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STAT 3202 · Chapter revision

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THE IDEA TO KEEP

Start with the question, then collect trustworthy data.

REMEMBER

A population is the full group; a sample is the part observed. A statistic estimates a population parameter.

APPLY IT

Measuring 30 randomly selected plots estimates yield for the field; measuring only the best plots introduces bias.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

1 What is statistics?

Statistics is the science which deals with the **collection, organisation (classification), presentation, analysis and interpretation of data**.

In agriculture: compare treatments, estimate yields and make decisions while accounting for variation and uncertainty.

2 Two branches

Statistics

Descriptive	Inferential
Summarises a sample or population with measures such as the mean or standard deviation.	Uses a random sample to describe and draw inferences about population parameters.

Data are individual pieces of factual information: the raw material of statistics (e.g. rainfall over a period, area under paddy in a state).

3 Population, sample, parameter

Population

The entire collection of units with the characteristic under study.

e.g. heights of **all** students in a college

Sample

A subset selected for study; should be **representative** of the population.

e.g. heights of 20 selected students

Parameter

A numerical characteristic of a population. It is **fixed** but usually **unknown**.

$\mu,\ \sigma^2,\ P$

Statistic

A value calculated from sample observations; used to **estimate** the parameter.

$\bar{x},\ s^2,\ \hat{p}$

4 Variables & constants

Variable: takes different values for different individuals, times or places. **Constant:** stays fixed (e.g. $\pi = 3.14159\ldots$, $e = 2.71828\ldots$).

Type	Meaning	Agri examples
Discrete quantitative	Countable, separate values (0, 1, 2 ...)	Fruits per branch, plots in a field
Continuous quantitative	Any value in a range	Plant height, yield, panicle length
Categorical qualitative	Categories, not measurements; can be counted	Variety, colour, disease present/absent

5 Measurement scales

Scale	Order	Equal gaps	True zero	Example
Nominal	✗	✗	✗	Blood group, gender
Ordinal	✓	✗	✗	Grades (excellent, good ...)
Interval	✓	✓	✗	Temperature in °C
Ratio	✓	✓	✓	Weight, Kelvin

40 °C is **not** "twice as hot" as 20 °C (no true zero), but 60 kg **is** twice 30 kg.

6 Collection of data

Primary data
First-hand, collected for the study's own purpose; collected specifically for the current investigation.
e.g. population census by the government

Secondary data
Originally collected for another purpose; reused in the current investigation.
e.g. Economic Survey, Ministry reports

PRIMARY METHODS personal investigation · trained enumerators · correspondents · phone or mailed questionnaire · online forms · mobile apps and social media polls

SECONDARY SOURCES official (RBI, ministries, NSO) · institutional (universities, research institutes). **Enumerator** collects; **respondent** provides.

7 Frequency distribution terms

Class limits	End points of a class, e.g. 110–119
Class boundaries	Close the gaps: 109.5–119.5 for data recorded to the nearest cm
Continuous classes	No gaps (10–20, 20–30): limits = boundaries; use $10 \le x < 20$
Class mark	Mid-point of the class interval
Class width	Upper boundary − lower boundary
LCF / GCF	Less-than (at upper boundaries) / greater-than (at lower boundaries) running totals
Relative freq.	f/N; all of them add up to 1

8 Building a grouped table

1. Decide the number of classes k
2. Determine the class width
3. Set up the class limits
4. Tally items into classes (every 5th stroke crosses the first four)
5. Count the frequency in each class

Raw data → **ungrouped** table (each value) or **grouped** table (classes). Grouping gives a clear picture but loses individual values.

9 Worked example · leaf length (cm), N = 38

Sturges: $k = 1 + 3.322 \log_{10} 38 \approx 6.25 \Rightarrow 6$ classes; width $(18-1)/6 \approx 2.83 \Rightarrow 3$ cm.

Class	Boundaries	Mark	f	LCF	GCF	RF
1–3	0.5–3.5	2	3	3	38	0.079
4–6	3.5–6.5	5	6	9	35	0.158
7–9	6.5–9.5	8	10	19	29	0.263
10–12	9.5–12.5	11	5	24	19	0.132
13–15	12.5–15.5	14	5	29	14	0.132
16–18	15.5–18.5	17	9	38	9	0.237

10 Formulae to remember

Number of classes (Sturges)
$$k = 1 + 3.322 \log_{10} N$$

Approximate class width
$$C = \frac{|\max - \min|}{k}$$

Lower boundary of class i
$$L_i = \frac{\text{LCL}_i + \text{UCL}_{i-1}}{2}$$

Class mark
$$\frac{\text{Lower limit} + \text{Upper limit}}{2}$$

Relative frequency
$$\text{RF} = \frac{f}{N}, \quad \sum \frac{f}{N} = 1$$

Sum of frequencies
$$\sum f = N$$

LCL / UCL = lower / upper class limit. Sturges' k is a guide, not a compulsory number.

! Common mistakes

- ✗ A large dataset cannot correct biased selection or poor measurement.
- ✗ Association alone does not prove cause and effect.
- ✗ Online forms and social media polls collect voluntary responses, so they are not automatically representative.
- ✗ Classes must cover the data **without overlap**. Boundaries depend on the recording precision.

? Fill in the blanks

1. Weight of a harvested mango recorded as 152.4 g is a ____ variable. **[continuous]**
2. The average yield of 100 sampled paddy farms is a ____; the true average of every farm in the district is a ____. **[statistic; parameter]**
3. Temperature in °C is on an ____ scale, so a 40 °C day is not "twice as hot" as a 20 °C day. **[interval]**
4. Rating farmers' satisfaction as low, medium or high uses the ____ scale. **[ordinal]**
5. A researcher using rainfall figures copied from an IMD report is working with ____ data. **[secondary]**
6. Plant height recorded to the nearest cm in the class 110–119 has class boundaries ____ and ____. **[109.5 and 119.5]**